DEPARTMENT 1 · DEVELOPMENT AND IT · PROFESSION 14

Data quality specialist

Profession 14 of 10 in Development and IT.

What the profession means

A data quality specialist defines and checks the rules that data must meet, measures how well it meets them, and works with owners to fix the causes.

What professionals do every day

Writes validation rules, runs checks on schedules, reports quality scores, and tracks issues to resolution.

Why businesses hire them, and what problems they solve

Bad data quietly corrupts every report built on it. Quality checks catch problems early and show where they come from.

- Problems found only by angry users.
- Rules that differ between teams.
- Fixes that treat symptoms.

Skills you need

- Validation rules: completeness, uniqueness, validity.
- SQL and Python checks.
- Root-cause analysis.
- Reporting quality scores.

Tools (free and paid)

Tool	What it is for	Free option	Paid option	Official link (not checked live)
SQLite	Learn SQL on a local file	Free	Free	www.sqlite.org
Python and pandas	Analysis in code	Free	Free	pandas.pydata.org
Great Expectations	Data validation framework (open source)	Free	Paid cloud	greatexpectations.io

Learning sequence: beginner, intermediate, advanced

Level	What you can do when you finish
Beginner	Write three rules for an orders table.
Intermediate	Run the rules and report pass or fail.
Advanced	Trace one failure to its cause.

Projects by level

- **Beginner:** Quality checks for a shop's orders.
- **Intermediate:** Customer data audit for a firm.
- **Advanced:** Survey response validation.

Client-ready portfolio project

A quality report for a fictional shop's orders: three rules, results, a failure traced to its cause and a fix plan.

Worked example

Worked example. Rule: every order has a positive amount. The check passes on the sample data; a row with amount 0 would fail, and the report traces it to the form that allowed a zero.

Assignment with full solution

ASSIGNMENT

Three rules and a result

Business problem: A shop wants its order data checked each week.

Steps

1. Write rules: amount is positive; customer id exists; order date is present.
2. Run each rule as a SQL check.
3. Report pass or fail with the count of rows.
4. Record one fix for any failure.

Expected output: The rules, the SQL and a pass report.

SOLUTION (ANSWER KEY)

SQL

```
-- data quality check: orders with a missing or non-positive amount
SELECT id, amount FROM orders WHERE amount IS NULL OR amount <= 0;
```

The first rule returns no rows on the sample data, so it passes. The customer rule and date rule are written the same way; any row returned is a failure to fix.

Common mistakes

- Rules with no owner.
- Checking once and never again.
- Fixing rows by hand without fixing the cause.
- No quality target.

Freelancing opportunities

- Data audits and quality programmes.
- Validation for analytics teams.
- Briefs for 'data quality', 'data validation', 'data audit'.

Learning resources

LEARNING RESOURCES (VERIFY BEFORE USE)

- Great Expectations documentation

Part of **Master Freelancing: Data Analytics and Data Science**. Back to the chapter list on skillnavora.in · Open the full textbook. Official links in this chapter were not checked live; confirm before relying on them.